

AirSense

Campus Pilot Initiative

Rooftop Air Quality Monitoring, Intelligent Forecasting and Campus Decision Support

A Business Proposal for Budget Approval, Hardware Deployment and Strategic Pilot Validation

SUBMITTED TO

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PKR 20,000 Budget Requested	3 Months Pilot Duration	2 Sites Islamabad + Karachi	4 Horizons 1h / 3h / 6h / 24h Forecasts
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1. Executive Summary

AirSense is an intelligent campus environmental decision platform developed by students at **Beaconhouse International College (BIC), Islamabad**. Recognized in international innovation competitions, AirSense transforms environmental data into direct, actionable business and administrative guidance for campus leadership.

This proposal requests formal approval for a one-time hardware investment of **PKR 20,000** to execute Phase 1 of the AirSense Campus Pilot at BIC Islamabad. The pilot includes installing a low-cost rooftop sensor unit, collecting three months of continuous air quality data, and calibrating predictive forecasting models. Operating in parallel with a sister pilot at BIC Karachi, this initiative creates Pakistan's first multi-campus smart environmental network.

Strategic Financial Assurance: The PKR 20,000 budget covers 100% of hardware procurement, installation, weatherproofing, and contingency reserves. **Zero cloud, subscription, or recurring costs** are incurred during the pilot. The institute's exposure is strictly limited to a single capital expenditure yielding lasting institutional assets.

At the conclusion of the pilot, BIC will possess a validated environmental dataset, calibrated forecasting engines, an operational decision playbook, and a published pilot report. These assets deliver clear brand leadership, student health protection, and proprietary intellectual property for Beaconhouse.

2. Strategic Background and Origin

2.1 The Origin of AirSense: Market-Validated Innovation

AirSense was conceived during an international academic innovation challenge, earning top recognition for its unique business focus. While traditional air quality systems output raw, complex graphs, AirSense operates at the **executive decision layer**: converting complex predictions into clear administrative actions such as adjusting building ventilation, rescheduling outdoor sports, or issuing student health advisories.

Early prototypes proved the core concept using public data. Moving to a live campus pilot provides the hyper-local microclimate data required to achieve commercial predictive accuracy.

2.2 Why BIC Islamabad is the Ideal Flagship Hub

Every successful commercial technology requires a controlled environment to validate operational utility. BIC Islamabad offers the ideal flagship location: clear physical boundaries, established power and network infrastructure, an active student community, and progressive institutional leadership.

By hosting this pilot, BIC Islamabad becomes the founding innovation hub for smart campus environmental management across the Beaconhouse network, generating the institution's first continuous local air quality dataset.

2.3 The Business, Health, and Operational Problem Addressed

Seasonal smog and fine particulate pollution (PM2.5) in the Islamabad region create significant operational challenges for educational institutions. Poor air quality directly affects student health, classroom focus, and attendance. Currently, campus administrators make operational decisions without predictive intelligence, often reacting too late to pollution spikes.

AirSense fills this gap by delivering 24-hour predictive risk forecasts, enabling BIC management to fulfill its duty of care for student health while optimizing campus facility operations.

3. The AirSense System

3.1 System Architecture: Transforming Data into Business Value

AirSense connects physical hardware to administrative decision-making through four simple layers:

System Layer	Hardware & Software Stack	Business & Institutional Value
Sensing Layer	PMS7003 + BME280 + Rain Sensors	Captures continuous rooftop particulate, temperature, humidity, and rain metrics.
Control & Logging	ESP32 Controller + Onboard MicroSD	Ensures 100% data security and continuous logging with zero external cloud dependencies.
Forecasting Engine	Machine Learning Algorithms	Generates 1-hour, 3-hour, 6-hour, and 24-hour ahead pollution risk predictions.
Decision Layer	Operational Playbook + AI Advisor	Translates predictions into plain-language administrative instructions for campus managers.

3.2 What Makes AirSense Different: The Decision Advantage

Conventional environmental monitors only show past pollution data. AirSense provides **predictive operational recommendations**. Rather than asking campus staff to interpret technical data, the platform provides direct advice: *'High smog expected in 4 hours. Switch HVAC intake to internal recirculation and move physical education classes indoors.'*

Furthermore, the platform is designed for rapid expansion across additional Beaconhouse campuses, offering a standardized blueprint for group-wide adoption.

3.3 Operational Capabilities and Input Drivers

The system combines multiple inputs to deliver accurate forecasts tailored to campus needs:

Category	Input Variable	Primary Source	Operational Benefit
Target Metric	PM2.5 Concentration	Rooftop Sensor	Core indicator used to trigger health warnings.
Historical Trends	Hourly Lag Records	Onboard Storage	Captures daily microclimate and traffic patterns.
Meteorology	Temperature & Humidity	BME280 Sensor	Models atmospheric conditions driving smog formation.
Rain Events	Rainfall Indicator	Rain Sensor	Identifies natural air cleansing events.
Wind Patterns	Wind Speed & Direction	Weather API	Forecasts regional pollution movement toward campus.
Campus Schedule	Academic Calendar Flags	System Calendar	Adjusts predictions based on term times and breaks.

4. Campus Pilot Implementation Plan

4.1 Strategic Pilot Objectives

- 1. Hardware Commissioning:** Deploy a weatherproofed rooftop sensor unit at BIC Islamabad to verify continuous operation.
- 2. Proprietary Data Asset:** Build a clean three-month environmental dataset owned entirely by BIC.
- 3. Model Accuracy Validation:** Calibrate machine learning models for 24-hour ahead air quality forecasting.
- 4. Administrative Utility Testing:** Evaluate automated administrative alerts against real campus events.

4.2 Phased Deployment Roadmap

The project follows a structured, risk-managed 6-phase roadmap designed for smooth execution without administrative burden:

Phase & Timeline	Activity Focus	Executive & Operational Deliverables
Phase 0 Days 1 to 7	Hardware Procurement	- Procure verified sensor components from local vendors. - Secure weatherproofing enclosure and power leads. - Perform initial quality checks before bench assembly.
Phase 1 Days 6 to 9	Bench Assembly & Benchmarking	- Assemble controller, sensors, and storage modules. - Upload system firmware and test data logging. - Confirm Wi-Fi data transmission to campus workstation.
Phase 2 Days 9 to 10	Rooftop Installation	- Mount weatherproof enclosure on BIC rooftop. - Connect power and verify Wi-Fi signal strength. - Initiate live data logging. Pilot clock starts.
Phase 3 Months 1 to 3	Autonomous Data Collection	- Sensor operates continuously with zero daily maintenance. - Data collection runs through weekends and academic breaks. - Weekly quality checks ensure uninterrupted telemetry.
Phase 4 Months 2 to 3	Model Training & Calibration	- Clean environmental dataset and remove anomalies. - Train time-series machine learning models. - Evaluate forecast accuracy across prediction horizons.

Phase & Timeline	Activity Focus	Executive & Operational Deliverables
Phase 5 End Month 3	Executive Reporting & Handover	- Evaluate operational decision accuracy against campus conditions. - Deliver comprehensive Pilot Completion Report to BIC Management. - Archive dataset and trained models; present expansion options.

Autonomous, Maintenance-Free Operation: Once installed, the rooftop unit runs autonomously without requiring daily staff attention. Data collection continues smoothly across academic breaks, building a robust asset for the college.

5. Multi-Campus Network Synergy: Karachi Collaboration

5.1 Two-City Pilot Network Strategy

Alongside the Islamabad deployment, a parallel pilot is coordinated at a Beaconhouse campus in Karachi led by a dedicated student lead. Utilizing the exact same hardware configuration and software pipeline, this joint effort creates Pakistan's first multi-campus student air quality network.

BIC Islamabad (Flagship Hub)	BIC Karachi (Coastal Sister Site)
- Primary development hub and baseline training site. - Inland climate: cold winter smog, temperature inversions, urban traffic. - Overseen directly by BIC Islamabad management.	- Parallel deployment validating cross-city compatibility. - Coastal climate: high humidity, sea breeze dispersion, industrial dust. - Demonstrates scalability across different regional environments.

5.2 Strategic Value to Beaconhouse

Testing across two distinct climate zones proves that AirSense can scale across the entire Beaconhouse Group. This multi-city dataset enhances predictive precision and positions BIC as a pioneer in campus technology.

6. Core Business Benefits to Beaconhouse International College

The AirSense pilot offers strong strategic, marketing, academic, and financial returns for BIC management:

Strategic ROI Pillar	Direct Institutional Value and Persuasive Business Impact
Duty of Care & Student Safety	Protects student and staff health with 24-hour predictive smog alerts. Minimizes health risks, reduces absenteeism, and demonstrates institutional responsibility.
Marketing & Admissions PR	Establishes BIC as <i>'Pakistan's First AI-Powered Smart & Healthy Campus'</i> . Provides a unique, premium story for student recruitment, open days, and social media campaigns.
Academic Distinction & Partners	Offers clear evidence of applied engineering leadership to UK degree awarding partners, proving BIC students build deployed, real-world solutions.
Facility & HVAC Cost Savings	Enables smart management of building ventilation and outdoor activities, avoiding energy waste and extending air filter life during high-pollution periods.
Departmental Prestige	Elevates the standing of the CSSE and AI Department across the Beaconhouse Group, proving practical innovation leadership.
100% Asset & IP Ownership	BIC retains full ownership of all hardware, datasets, calibrated models, and research outputs for future academic use or commercial scaling.

7. Executive Control Dashboard Overview

The AirSense dashboard provides campus leadership with an intuitive interface showing live air metrics, forecast trends, regulatory notices, and conversational AI guidance.

7.1 Main Executive Dashboard and Policy Tracker

The primary dashboard displays real-time air quality metrics, 10-day risk trends, and regulatory alerts to support quick administrative decision-making.

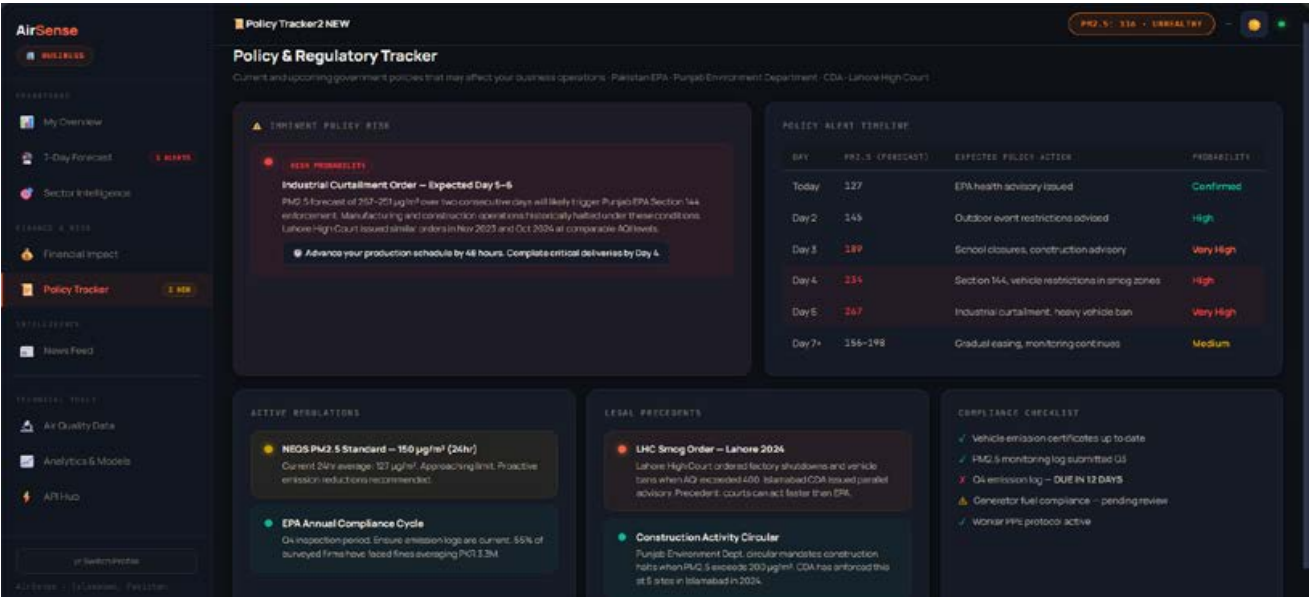


Figure 7.1a: AirSense Main Overview displaying live air metrics and predictive risk horizons.

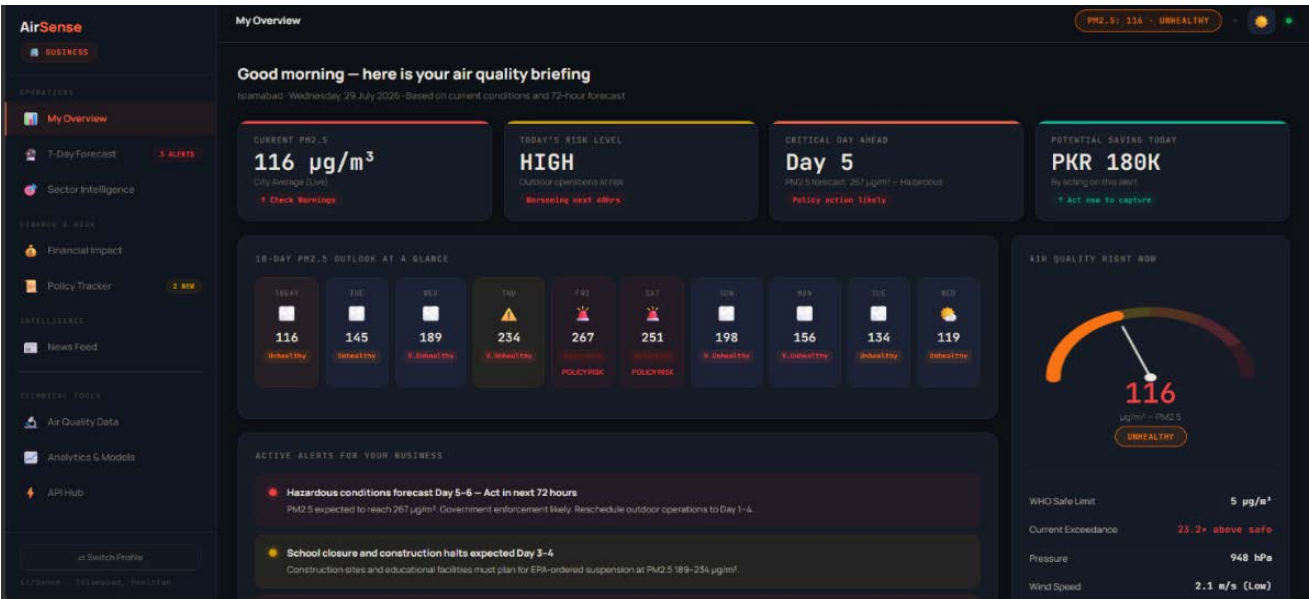


Figure 7.1b: Regulatory Tracker monitoring environmental directives and campus compliance.

7.2 AI Decision Advisor and Health Risk Calculator

Natural language AI interfaces allow administrators to ask operational questions, while the health calculator recommends safe activity windows for campus groups.

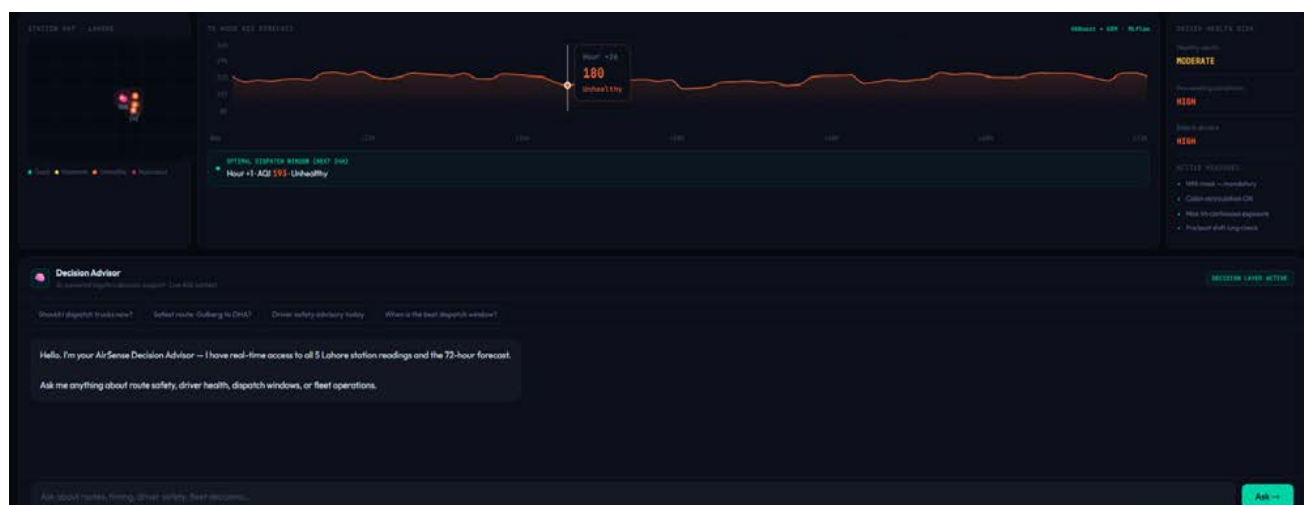


Figure 7.2a: AI Advisor providing plain-language operational guidance to campus leadership.

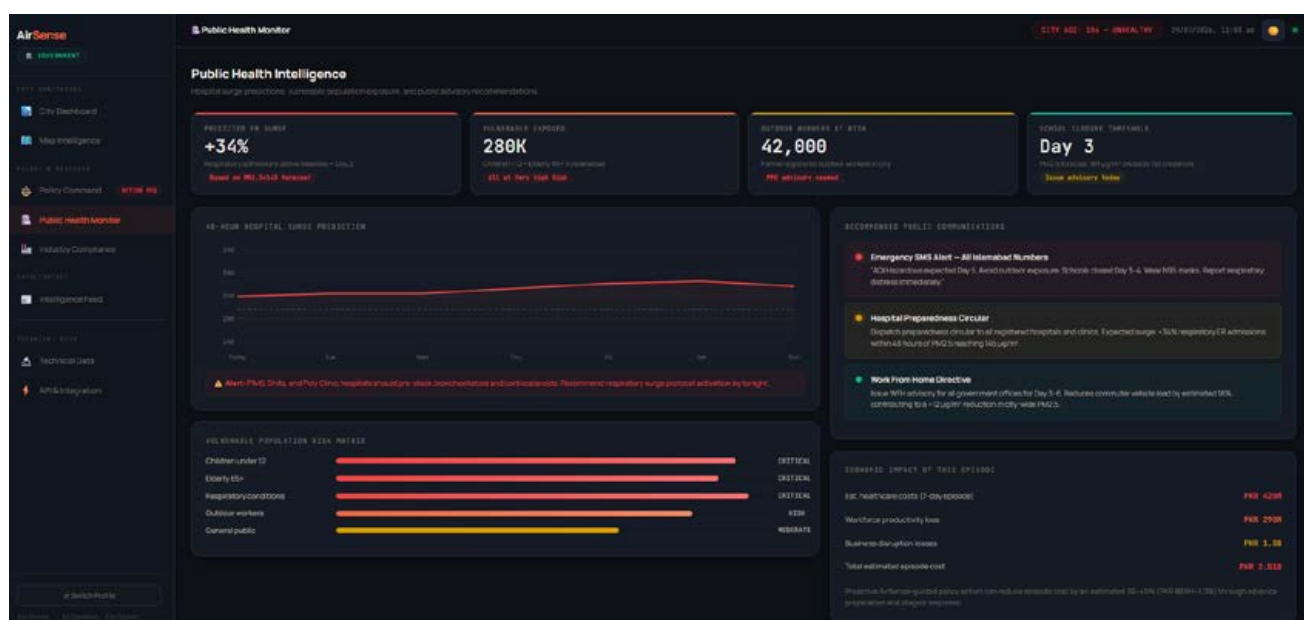


Figure 7.2b: Health Risk Panel calculating safe dispatch windows for outdoor activities.

7.3 Policy Command Centre

The Policy Command Centre coordinates campus emergency responses during high pollution episodes, issuing automated notices to facility managers.

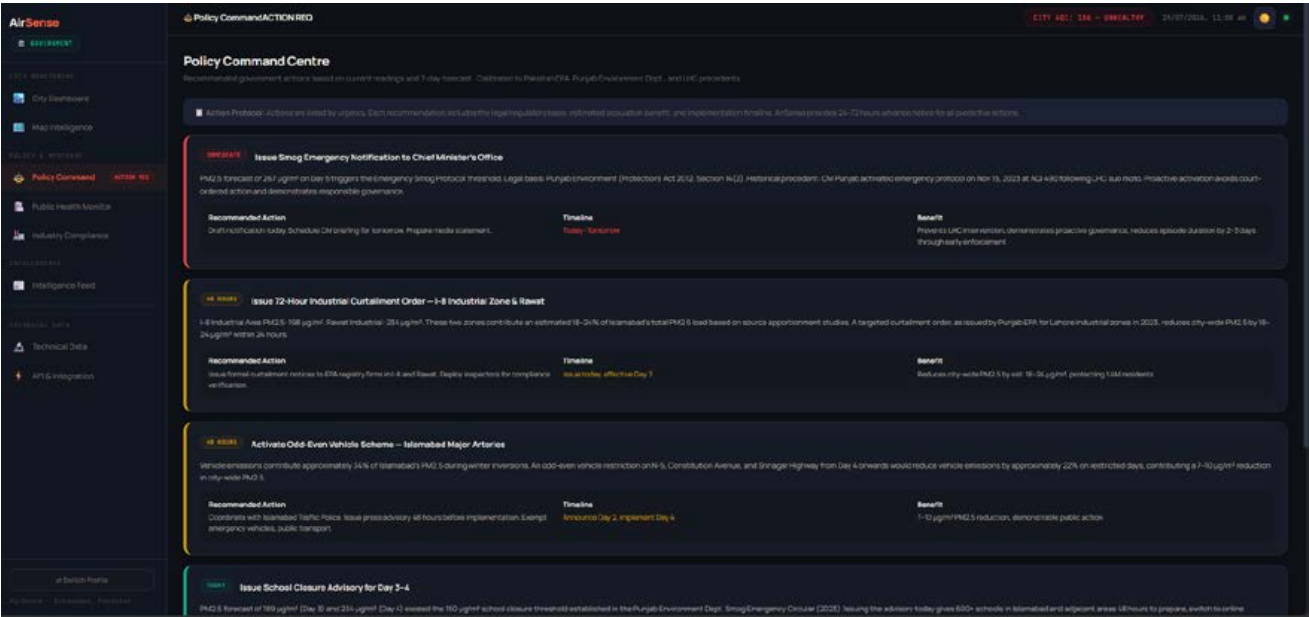


Figure 7.3: Policy Command Centre mapping automated advisory triggers to facility actions.

8. Commercial Vision and Expansion Scope

8.1 From Campus Pilot to Commercial SaaS Platform

Phase 1 at BIC Islamabad establishes the foundational data and operational blueprint for a scalable commercial SaaS business targeting enterprise clients.

Phase	Growth Stage	Strategic Scope and Objectives	Key Institutional Deliverable
1	Campus Pilot	Deploy rooftop units at BIC Islamabad and Karachi. Collect 3 months data to train baseline prediction engines.	Validated local dataset, calibrated models, pilot completion report.
2	Network Expansion	Refine algorithms and expand deployment across additional Beaconhouse Group campuses nationwide.	High-accuracy multi-campus model suite, standardized blueprint.
3	Commercial Launch	Launch enterprise SaaS product targeting logistics, construction, healthcare, and educational sectors.	Revenue-generating commercial platform with BIC as founding IP partner.

8.2 B2B Enterprise Market Opportunities

AirSense addresses clear operational needs across multiple high-value commercial markets:

Enterprise Sector	Environmental Trigger	Commercial Value Proposition
Logistics & Transport	High morning smog forecast	Adjust shift departure times and optimize routes to reduce driver exposure and delivery delays.
Schools & Campuses	PM2.5 spike in next 3 to 6 hours	Move sports activities indoors and adjust building ventilation to protect student health.
Construction & Mining	High pollution + low wind dispersion	Activate dust suppression misting and issue safety gear to outdoor work crews.
Retail & Hospitality	Severe smog episode alert	Optimize indoor air purification and highlight clean air environments to guests.
Corporate ESG Reporting	Repeated threshold exceedance	Generate automated compliance reports for corporate sustainability audits.

BIC as Founding Technology Partner: Approving this pilot establishes BIC Islamabad as the founding institutional partner of AirSense, creating lasting academic recognition and commercial licensing potential.

9. Executive Budget Request

9.1 Itemized Hardware Budget (PKR 20,000)

The requested **PKR 20,000** represents a single, one-time capital investment covering all physical components, enclosures, weatherproofing, and power leads. Zero recurring costs are required.

#	Component Description	Vendor Source	Qty	Cost (PKR)
1	PMS7003 Laser Dust PM1/PM2.5/PM10 Sensor	Embeded Studio	1	4,200
2	ESP32 WROOM-32D Development Board	Digilog.pk	1	1,160
3	BME280 Temp / Humidity / Pressure Sensor	Electrobes	1	850

#	Component Description	Vendor Source	Qty	Cost (PKR)
4	Arduino MicroSD Card Reader Module	Electrobes	1	150
5	Samsung EVO Plus 16 GB MicroSD Card	Daraz.pk	1	1,450
6	Waterproof IP65 Electrical Enclosure Box	A.E Solution	1	750
7	Jumper Wire Kit, 120 pcs	Daraz.pk	1	440
8	5V 2A AC/DC Power Adapter Unit	Electronics Hub	1	170
9	Cable Ties, 100-pack	Daraz.pk	1	120
10	2-inch Stainless Hose Clamps (2 pcs)	DreamsMart.pk	1	60
11	Raindrop Detection Sensor Module A	Digilog.pk	1	160
12	Rain Drop Moisture Sensor Module B	Electrobes	1	150
13	UNI-T UT363 Digital Anemometer	Electrobes	1	4,250
14	Clopal 10m Heavy-Duty Extension Lead	Clopal Online	1	3,295
15	GMSA RTV Silicone Weatherproof Sealant	Expert Tools	1	390
16	Mounting Nuts & Bolts M3-M6 (8 pcs)	Multan Elec.	8	192
17	Radiation Shield (Local Islamabad Sourcing)	Local Supplier	1	TBD
	Hardware Subtotal (Items 1 to 16)			PKR 17,787
	Radiation Shield + Contingency Reserve Buffer			PKR 2,213
	TOTAL BUDGET REQUESTED			PKR 20,000

Financial Governance: The radiation shield price is pending final local quotation. The PKR 2,213 contingency buffer is reserved to cover this component and minor price fluctuations. Any unspent funds will be returned to the institute.

9.2 Financial Sustainability

Phase 2 operates with zero additional budget by utilizing the Phase 1 hardware stack. Commercial SaaS costs in Phase 3 will be self-funded through client revenue.

10. Deliverables at Pilot Completion

At the end of the 3-month pilot, the project team will formally submit five core deliverables to BIC management:

No.	Deliverable Asset	Institutional Description and Strategic Value
1	Validated PM2.5 Dataset	3 months of continuous, timestamped hourly air quality records collected directly from the BIC Islamabad campus.
2	Trained Forecasting Models	Calibrated machine learning models for 1h, 3h, 6h, and 24h prediction horizons with performance metrics.
3	Decision Framework Report	An evaluation of the AI decision layer: testing advice accuracy and real-world administrative utility on campus.
4	Environmental Drivers Analysis	An analytical report identifying the primary atmospheric drivers of local Islamabad pollution.
5	Pilot Completion Report	A comprehensive executive report detailing system performance, multi-campus findings, and expansion options.

11. Conclusion and Formal Request for Approval

AirSense represents an exceptional, low-risk opportunity for Beaconhouse International College. For a modest one-time investment of **PKR 20,000**, BIC gains a functional smart environmental station, proprietary data assets, calibrated AI decision engines, and a unique marketing asset that reinforces BIC's reputation as a leader in educational innovation.

The project team is ready to commence deployment immediately upon approval.

Formal Request for Management Approval

The AirSense team respectfully requests approval of the following items from **Ms. Saba Ahson** and **Ms. Sahifa Alam**:

1	Approval of a one-time hardware budget of PKR 20,000 for procurement of sensor components itemized in Section 9.
2	Permission to install the weatherproofed sensor unit at an agreed rooftop location on the BIC Islamabad campus for three months.
3	Facilitation of access to a rooftop power outlet and setup of a campus Wi-Fi extender for live telemetry.
4	Official recognition as a supported BIC student project, enabling access to campus computing resources for model training.

The sensor goes on the roof. The data comes in. The models learn.
We ask for PKR 20,000 and three months to prove that it works.
Everything else follows from that.

Respectfully submitted,

Munim Qureshi
AirSense Project Lead
Beaconhouse International College, Islamabad
July 2026

Approval Sign-Off

Role	Name	Signature	Date
Head of Institute	Ms. Saba Ahson		
Head of CSSE / AI	Ms. Sahifa Alam		
Project Team Lead, Islamabad	Munim Qureshi	Munim	29/07/2026
Project Team Lead, Karachi	Areesha Aqeel		